

T-Shape Microstrip Antenna for Dual-Band Operation

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Abstract

For the demands of multiple functions and integration with circuits for commercial communication systems, this paper uses the concept of electric field distortion to design a simple and coplanar T-shape microstrip antenna for dual-band operation. The design concept is to cut a conventional metallic patch into T shape, and to excite higher and lower modes within two different resonant cavities. With specific dimensions of an antenna, the radiation patterns of these two modes are acceptable, and their impedance is easy to match. In the theoretical analysis, the cavity model is used to first divide the T-shape patch into two rectangular cavities, and then Galerkin's moment method is employed to obtain antenna parameters, such as resonant frequencies, electric fields and radiation patterns. The agreement between theoretical, simulated and experimental results confirms with design concept. The physical-intuitive analysis makes the antenna mechanism much easier to understand. Since shape cutting is the major idea, a microstrip antenna could achieve multi-band operation if the cutting number increases.

Keywords: Dual-Band, T-shape, Microstrip Antenna, Field Distortion, Galerkin's moment method

1. Background and Motivation

Recently, wireless commercial communication products have been so popular that the demands of antennas also increase. Therefore, how to design a compact and multi-functional antenna becomes an important issue. Among those types of antennas, microstrip antennas have been paid attention due to their advantages of compactness and system integration ability. However, the conventional microstrip antenna has only one mode for use [1]; therefore, there are many researches which are working on wide band or multiple-mode operation. For example, a rectangular microstrip with U shape or a parallel concave can achieve the goal for wide band application [2-3]. A microstrip antenna with

a F-shape feeding can also produce the similar purpose [4-5]. Another design is to reshape the rectangular patch into a diamond shape or a combination of four triangular shapes with improved feeding can also provide the characteristic of wide band [6]. However, among these design, it is difficult to maintain an acceptable radiation over such a wide range of frequency band. For this reason, a design with a discrete type of dual-band or triple-band antennas has been earned attention too. For example, a conventional microstrip antenna with PEC (perfect electric conductor) walls can easily approach dual-band operation [7]. The adjustment of feeding location can vary the separation of two modes, but the cross-polarization radiation of the first mode needs to improve. Another way is to add electromagnetic bandgap with a dielectric substrate [8]; however, the radiation is not sufficiently good. A microstrip with slot holes and chip capacitors is another design to offer dual band operation [9]. However, the lumped component is adequate in RF application. An antenna with V-type slot can also provide similar function, and this design is to distract those resonant lengths to create two modes [10]. Among the design mentioned here, adding passive components or slots on microstrip can really work, but the simple cavity of antenna has been also destroyed, such that the analysis or characteristic adjustment will become much more difficult for further design.

On the other hand, a microstrip antenna with conductive VIA around its sides can create two bands [11]. Adding air gap within a substrate can make the impedances of two modes easy to match. However, the non-coplanar structure limits the application of this design. An antenna with disturbing patches can be used to distract simple electric fields into two different distributions of modes. However, the disturbing effect is not easy to handle [12]. Therefore, how to use an easy concept to design a simple and coplanar antenna for dual band operation is the main purpose of this paper.

This paper applies the concept of cutting a conventional rectangular patch into a T-shape structure as shown in Figure 1. Due to the change of patch shape, two resonant modes are created with two different lengths. For the lower mode, the length of the long cavity is half wavelength, while the short one creates the higher mode. In addition, the radiations of these two modes can be adjusted by using the magnetic currents on the side walls of the cavity in order to make sure that the radiations are acceptable. For the lower mode, the magnetic

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current along the long side are in phase, and the one along the short side is not even, such that the radiation with a single beam is produced due to the construction interference. On the other hand, there are in-phase magnetic currents along the short side for the higher mode, and the magnetic currents along the long side are out of phase. Fortunately, the area for the out-of-phase one is so small that the overall radiation would not be affected too much. Therefore, the radiation of the higher mode can remain single beamed. In other words, the T-shape microstrip antenna can work at dual bands.

In order to confirm design concept, the choice of verification data is very important. Since the idea is based on the distortion of electric fields in a cavity, only experiments and simulations will not reveal the mechanism inside the antenna; therefore, another job of this paper is to provide theoretical results to complete the analysis.

From the cavity model point of view, the side-wall boundaries of this antenna are so ensured, that the whole antenna can be treated as a cavity [13]. However, this structure is not just a simple rectangular cavity, and the electric fields inside are more complicated. For this reason, this paper divides the T-shape cavity into two rectangular ones which are actually connected together by a conjunction. Therefore, how to use the continuity of electromagnetic waves over this conjunction to derive the antenna parameters is the major work for theoretical analysis. Since this kind of problem has been solved in related researches earlier [14-17], this paper will follow similar procedures to deal with the T-shape cavity here. Meanwhile, the technique of moment method will be employed to find antenna parameters. Further, the electric fields on PMC (perfect magnetic conductor) walls are converted into magnetic currents for radiation computation. The details are described in Section II.

2. Theoretical Formulation

For the T-shape microstrip antenna in Figure 1, the structure includes a T-shape patch, a dielectric substrate and an infinite metallic ground. This antenna is fed by a coaxial cable. According to the cavity model concept, the patch and the ground are treated as PEC, and the surrounding side walls are PMC, as shown in Figure 1. Since the geometry of this patch is a T shape, the cavity structure can be divided into regions I and II as indicated in Figure 2. Since the right and lower sides of cavity I and left and lower sides of cavity II are PMC, their individual vector magnetic potentials along x and y are functions of cosine harmonics. The substrate is so thin that these potentials are uniformly distributed along z . Therefore, the lowest mode of this antenna should be TM mode. The vector magnetic potentials in two regions are expressed as:

In region I,

$$\Psi_1 = \sum_{n=0}^N A_n \cos[k_{x1}(x-l)] \cos[k_{y1}(y-c)] \quad (1a)$$

$$k_{x1} = \sqrt{k^2 - k_{y1}^2} \quad (1b)$$

$$k_{y1} = \frac{n\pi}{w} \quad (1c)$$

In region II,

$$\Psi_2 = \sum_{m=0}^M B_m \cos[k_{x2}(x+b)] \cos[k_{y2}y] \quad (2a)$$

$$k_{x2} = \sqrt{k^2 - k_{y2}^2} \quad (2b)$$

$$k_{y2} = \frac{m\pi}{c+w+d} \quad (2c)$$

Where A_n and B_m are the coefficients for specific harmonics, m, n are harmonic number, and $k_{x1}, k_{x2}, k_{y1}, k_{y2}$, are wavenumbers along various axis. Since the electromagnetic waves can be transformed from these vector magnetic potentials, they become:

$$E_{x1} = 0 \quad (3a)$$

$$H_{y2} = \frac{1}{\mu} \sum_{m=0}^M B_m \cos[k_{x2}(x+b)] \cos(k_{y2}y) \quad (4e)$$

Since the tangential components of electromagnetic waves should be continuous at a boundary, it also happens at the conjunction between two cavities. In other words, H_{y1} is equal to H_{y2} at $x=0$, and E_{z1} is equal to E_{z2} , so

$$H_{y1}(y)|_{x=0} = H_{y2}(y)|_{x=0} \quad (5)$$

$$E_{z1}(y)|_{x=0} = E_{z2}(y)|_{x=0} \quad (6)$$

Therefore, the relationship between two cavities is established. As the continuity of magnetic fields is used,

$$\begin{aligned} & \sum_{n=0}^N A_n \cos[k_{x1}(x-l)] \cos[k_{y1}(y-c)] \sin[k_{x1}l] \\ &= \sum_{m=0}^M B_m \cos[k_{x2}(x+b)] \cos(k_{y2}y) \sin(k_{x2}b) \end{aligned} \quad (7)$$

If one uses $h(y)$ to represent the magnetic field at $x=0$ over the conjunction:

$$h(y) = \sum_{n=0}^N A_n \cos[k_{y1}(y-c)] \quad (8a)$$

$$h(y) = \sum_{m=0}^M B_m \cos(k_{y2}y) \quad (8b)$$

Where A_n and B_m are

$$A_n = A_n \cos[k_{x1} \sin(k_{x1}l)] \quad (9a)$$

$$B_m = B_m \cos[k_{x2} \sin(k_{x2}b)] \quad (9b)$$

Further, along the location y within conjunction region, after using Fourier expansion,

$$A_n = \frac{1}{w} \int_c^{c+w} h(y) \cos[k_{y1}(y-c)] dy \quad (10a)$$

$$B_m = \frac{1}{c+w+d} \int_c^{c+w} h(y) \cos(k_{y2}y) dy \quad (10b)$$

Or

$$A_n = \frac{1}{w \cos[k_{x1} \sin(k_{x1}l)]} \int_c^{c+w} h(y) \cos[k_{y1}(y-c)] dy$$

$$B_m = \frac{1}{(c+w+d) \cos[k_{x2} \sin(k_{x2}b)]} \int_c^{c+w} h(y) \cos(k_{y2}y) dy$$

Similarly, as the continuity of electric fields at conjunction is employed, the equation becomes:

$$\begin{aligned} & \sum_{n=0}^N A_n \cos[k_{y1}(y-c)] \cos[k_{x1}l] \\ & - \sum_{m=0}^M B_m \cos(k_{y2}y) \cos(k_{x2}b) = 0 \end{aligned} \quad (12)$$

Although $h(y)$ in A_n and B_m is undetermined, it can be expanded into the combination of chosen basis functions and coefficients:

$$h(y) = \sum_{p=0}^P \alpha_p f_p(y) \quad (13)$$

Where f_p is the p -th-order basis function and α_p is the corresponding coefficient. After inserting (13) into (12), the equation can be rearranged as

$$\begin{aligned} & \sum_{n=0}^N \sum_{p=0}^P C_1 \int_c^{c+w} \left[\sum_{p=0}^P \alpha_p f_p(y) \right] \cos[k_{y1}(y-c)] dy \cos[k_{x1}l] \\ & - \sum_{m=0}^M \sum_{p=0}^P C_2 \int_c^{c+w} \left[\sum_{p=0}^P \alpha_p f_p(y) \right] \cos(k_{y2}y) dy \cos(k_{x2}b) = 0 \end{aligned}$$

Where C_1 and C_2 :

$$C_1 = \frac{\cos(k_{x1}l)}{w \cos[k_{x1} \sin(k_{x1}l)]} \quad (15a)$$

$$C_2 = \frac{\cos(k_{x2}b)}{(c+w+d) \cos[k_{x2} \sin(k_{x2}b)]} \quad (15b)$$

As the $h(y)$ expands,

$$\begin{aligned} & \sum_{n=0}^N C_1 \int_c^{c+w} (a_1 f_1 + a_2 f_2 + a_3 f_3 + \dots + a_p f_p) \cos[k_{y1}(y-c)] dy \cos[k_{x1}l] \\ & - \sum_{m=0}^M C_2 \int_c^{c+w} (a_1 f_1 + a_2 f_2 + a_3 f_3 + \dots + a_p f_p) \cos(k_{y2}y) dy \cos(k_{x2}b) = 0 \end{aligned}$$

According to Galerkin's moment method, if a testing function f_q is applied on the equation above, then

As $q=1$,

$$\begin{aligned} & \sum_{n=0}^N C_1 \int_c^{c+w} (a_1 f_1 + a_2 f_2 + a_3 f_3 + \dots + a_p f_p) \cos[k_{y1}(y-c)] dy \int_c^{c+w} f_1 \cos[k_{y1}(y-c)] dy \\ & - \sum_{m=0}^M C_2 \int_c^{c+w} (a_1 f_1 + a_2 f_2 + a_3 f_3 + \dots + a_p f_p) \cos(k_{y2}y) dy \int_c^{c+w} f_1 \cos(k_{y2}y) dy = 0 \end{aligned}$$

As $q=2$,

$$\begin{aligned} & \sum_{n=0}^N C_1 \int_c^{c+w} (a_1 f_1 + a_2 f_2 + a_3 f_3 + \dots + a_p f_p) \cos[k_{y1}(y-c)] dy \int_c^{c+w} f_2 \cos[k_{y1}(y-c)] dy \\ & - \sum_{m=0}^M C_2 \int_c^{c+w} (a_1 f_1 + a_2 f_2 + a_3 f_3 + \dots + a_p f_p) \cos(k_{y2}y) dy \int_c^{c+w} f_2 \cos(k_{y2}y) dy = 0 \end{aligned}$$

As $q=Q$,

$$\begin{aligned} & \sum_{n=0}^N C_1 \int_c^{c+w} (a_1 f_1 + a_2 f_2 + a_3 f_3 + \dots + a_p f_p) \cos[k_{y1}(y-c)] dy \int_c^{c+w} f_Q \cos[k_{y1}(y-c)] dy \\ & - \sum_{m=0}^M C_2 \int_c^{c+w} (a_1 f_1 + a_2 f_2 + a_3 f_3 + \dots + a_p f_p) \cos(k_{y2}y) dy \int_c^{c+w} f_Q \cos(k_{y2}y) dy = 0 \end{aligned}$$

As the same function as the basis function is chosen for testing, the equation can be arranged as a $[Z]_{p \times q}$ matrix:

$$\begin{bmatrix} z_{11} & z_{12} & \cdots & z_{1p} \\ z_{21} & z_{22} & \cdots & z_{2p} \\ \cdots & \cdots & z_{qp} & z_{3p} \\ z_{Q1} & z_{Q2} & \cdots & z_{QP} \end{bmatrix} \cdot \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \vdots \\ \alpha_p \end{bmatrix} = 0 \quad (18)$$

Where the matrix element,

$$z_{qp} = -\sum_{n=0}^N \frac{\cos(k_{x1}l)}{w[k_{x1}]\sin(k_{x1}l)} \langle f_p | f_q \rangle_{i=1} - \sum_{m=0}^M \frac{\cos(k_{x2}b)}{(c+w+d)[k_{x2}]\sin(k_{x2}b)} \langle f_p | f_q \rangle_{i=2} \quad (19)$$

Where the inner product $\langle f_p | f_q \rangle$ represents:

$$\langle f_p | f_q \rangle_{i=1} = \int_c^{c+w} f_p(y) \cos[k_{y1}(y-c)] dy \int_c^{c+w} f_q(y) \cos[k_{y1}(y-c)] dy$$

$$\langle f_p | f_q \rangle_{i=2} = \int_c^{c+w} f_p(y) \cos(k_{y2}y) dy \int_c^{c+w} f_q(y) \cos(k_{y2}y) dy$$

Since the coefficient $[\alpha]$ should exist, the determinant of Z matrix should be zero. Note that the element in [Z] includes the overall wavenumber of the whole cavity, and the roots of the matrix can be converted to resonant frequency of the T-shape microstrip antenna. As the obtained resonant frequency is put back to the dependent equations in (18), the ratio of coefficients for the chosen basis functions at specific resonant frequency can be found as well as A_n and B_m . After transforming the vector magnetic potentials in regions I and II, the electric fields along z axis become

$$E_{z1} = -j \frac{1}{\omega \mu \epsilon} k^2 \sum_{n=0}^N A_n [\cos[k_{y1}(y-c)] [\cos[k_{x1}(x-l)]]$$

$$E_{z2} = -j \frac{1}{\omega \mu \epsilon} \sum_{m=0}^M B_m [\cos(k_{y2}y) [\cos[k_{x2}(x+b)]] \quad (21b)$$

These electric fields can be used to computer other antenna parameters, such as magnetic currents, radiations.

According to the cavity model, the radiation is converted from the magnetic currents on PMC walls of a microstrip cavity. And the magnetic current distributions are indicated in Figure 3, where

$$\overline{M}_1 = 2 \sum_{n=0}^N A_n \cos[k_{x1}(x-l)] a_x \quad (22a)$$

$$\overline{M}_2 = -2 \sum_{n=0}^N A_n \cos(n\pi) \cos[k_{x1}(x-l)] a_x \quad (22b)$$

$$\overline{M}_3 = 2 \sum_{m=0}^M B_m \cos[k_{x2}(x+b)] a_x \quad (22c)$$

$$\overline{M}_4 = -2 \sum_{m=0}^M B_m \cos(m\pi) \cos[k_{x2}(x+b)] a_x \quad (22d)$$

$$\overline{M}_5 = -2 \sum_{m=0}^M B_m \cos(k_{y2}y) a_y \quad (22e)$$

$$\overline{M}_6 = 2 \sum_{m=0}^M B_m \cos(k_{y2}y) \cos(k_{x2}b) a_y \quad (22f)$$

$$\overline{M}_7 = \overline{M}_6 \quad (22g)$$

$$\overline{M}_8 = 2 \sum_{n=0}^N A_n \cos[k_{y1}(y-c)] a_y \quad (22h)$$

Among these magnetic currents, M_1 to M_4 are the one along x-axis, and M_5 to M_8 are along y-axis. Therefore, the vector electric potentials F_x and F_y of radiation become

$$F_x = \iint_s \frac{e^{-jk_0 r}}{4\pi r} (M_1 + M_2 + M_3 + M_4) e^{jk_0 r' \cos \phi} ds'$$

$$F_y = \iint_s \frac{e^{-jk_0 r}}{4\pi r} (M_5 + M_6 + M_7 + M_8) e^{jk_0 r' \cos \phi} ds'$$

Where the area for integration is the region of a PMC wall, and r and r' are the distances between observation point and origin or current sources. Note that $r' \cos(\phi)$ can be arranged as

$$r' \cos \phi = x' \sin \theta \cos \phi + y' \sin \theta \sin \phi + z' \cos \theta$$

For those radiating electric potentials, they become

$$F_x = F_1 + F_2 + F_3 + F_4 \quad (25a)$$

$$F_y = F_5 + F_6 + F_7 + F_8 \quad (25b)$$

$$F_1 = D \sum_{n=0}^N A_n e^{jk_0 [c] \sin \theta \sin \phi} \int_0^l \cos[k_{x1}(x-l)] e^{jk_0 [x] \sin \theta \cos \phi} dx$$

$$F_2 = D \sum_{n=0}^N A_n [\cos(n\pi) e^{jk_0 [(c+w)] \sin \theta \sin \phi}] \int_0^l \cos[k_{x1}(x-l)] e^{jk_0 [x] \sin \theta \cos \phi} dx$$

$$F_3 = D \sum_{m=0}^M B_m \int_{-b}^0 \cos[k_{x2}(x+b)] e^{jk_0 [x] \sin \theta \cos \phi} dx$$

$$F_4 = D \sum_{m=0}^M B_m \cos(m\pi) e^{jk_0 [(c+w+d)] \sin \theta \sin \phi} \int_{-b}^0 \cos[k_{x2}(x+b)] e^{jk_0 [x] \sin \theta \cos \phi} dx$$

$$F_5 = D \sum_{m=0}^M B_m e^{jk_0 [-b] \sin \theta \sin \phi} \int_0^{c+w+d} \cos(k_{y2}y) e^{jk_0 [y] \sin \theta \sin \phi} dy$$

$$F_6 = D \sum_{m=0}^M B_m \cos(k_{x2}b) \int_{c+w}^{c+w+d} \cos(k_{y2}y) e^{jk_0 [y] \sin \theta \sin \phi} dy$$

$$F_7 = D \sum_{m=0}^M B_m \cos(k_{x2}b) \int_0^c \cos(k_{y2}y) e^{jk_0[y] \sin \theta \sin \phi} dy$$

$$F_8 = D \sum_{n=0}^N A_n e^{jk_0[x] \sin \theta \cos \phi} \int_0^{c+w} \cos[k_{y1}(y-c)] e^{jk_0[y] \sin \theta \sin \phi} dy$$

Where

$$D = \frac{e^{-jk_0 r} [e^{jk_0 [x] \cos \theta} - 1]}{2\pi r jk_0 \cos \theta} \quad (27)$$

Finally, the far-field radiating electric fields are

$$E_r \approx 0 \quad (28a)$$

$$E_\theta = -jk_0 (-F_x \sin \phi + F_y \cos \phi) \quad (28b)$$

$$E_\phi = jk_0 (F_x \cos \theta \cos \phi + F_y \cos \theta \sin \phi) \quad (28c)$$

3. Results and Comparisons

This section uses the comparison between theory, simulations and experiments to verify the theoretical analysis for T-shape microstrip antenna. The simulated data is obtained from commercial software ENSEMBLE. For the antenna with specific dimensions, Figure 4 shows the S_{11} parameter vs. operating frequency. The resonant frequency of the lower mode from theory is 1.74 GHz, while those from simulations and experiments are 1.61 GHz. Note that the theoretical result is converted from the input impedance computed in [18]. The little difference shows the agreement between the results obtained from these three methods. On the other hand, the resonant frequency of higher modes from theory, simulations, and experiments are 2.88, 2.83 and 2.73 GHz. The only 5 percentage error also demonstrates the agreement between these data.

Figure 5 shows the resonant frequencies of the two modes vs. the number of used basis function on the conjunction of a T-shape microstrip antenna. As this number increases, the theoretical data converges only with enough basis functions. With various basis numbers, Figures 6 and 7 reveal the magnetic field $h(y)$ on the conjunction for both two modes. As this number is large enough, the $h(y)$ also converges.

The distribution of electric fields along the cross section of the antenna in Figure 1 is demonstrated in Figure 8. Again, the simulated result is transferred from the input impedance at the feeding location of this antenna. The one of the lower mode reveals the half-wave distribution, and the one of the higher mode is about the distribution of a whole wavelength. This result seems reasonable since the distribution for the higher mode is more

complicated than the lower one. Moreover, the amplitudes of the electric fields at both sides are even for the higher mode, so the strengths of corresponding magnetic currents are equal, but out of phase.

Figure 9 shows the electric field distribution along the upper cross section for the two modes. The quarter-wave and half-wave distributions are due to the distortion on T-shape structure. Although there is a limited error on simulated data, basically it agrees with those data from theory, so the theoretical result is verified. For the lower mode, the strength of electric fields is so weak that the magnetic current at the left side of antenna will dominate the whole antenna radiation. The distributions of electric fields along the vertical section are shown in Figure 10. Apparently, these distributions are roughly uniform, which it means the modes of interest are as expected.

According to the Figures 8 to 10, the magnetic currents of this T-shape antenna are shown in Figure 11. For the lower mode, those magnetic currents along y-axis are in phase, so that the one at the right side in region I will dominate antenna radiation, which is single beamed. However, those magnetic currents along x-axis are out of phase, so that the mutual destruction makes lower cross-polarization radiation. For the higher mode, although, the magnetic current at right side of region I points opposite direction with those at other locations, the overall co-polarization radiation still remains single beamed. While most of the magnetic currents along x-axis can cancel each other, the strength of cross-polarization radiation is pretty low. According these results, the radiations for the two modes should be acceptable.

After theoretical computation, the radiation patterns of these two modes are shown in Figures 12 and 13. The agreement between theory and simulation again confirm the theoretical analysis and design concept.

4. Conclusion

In order to produce a dual-band and circuit-integration antenna, this paper uses the concept of field distortion to design a coplanar T-shape microstrip antenna. Theoretical, simulated and experimental data all show that the electric fields are indeed distorted due to employing a cutting patch. Different lengths of resonant cavities can create lower and higher modes, and they both have acceptable radiation patterns. The theoretical analysis of this paper uses the cavity model to divide this T-shape antenna into two cavities. After using the moment method technique, electric fields, resonant frequencies and radiation patterns are obtained. With the valuable information, the design mechanism is much easier to understand. The

design concept is confirmed by theoretical analysis, and the validity of the theoretical analysis is also verified by simulations and experiments. The achievement of the dual band function for this T-shape antenna is achieved due to two resonant lengths on a patch. As the number of these lengths increases, an easy approach to provide multiple-band operation for a microstrip antenna becomes possible.

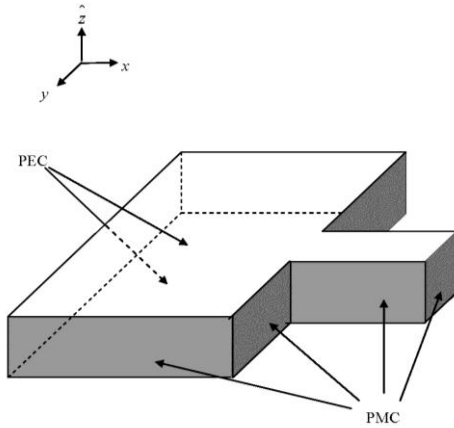


Figure 1: The cavity of the proposed T-shape microstrip antenna.

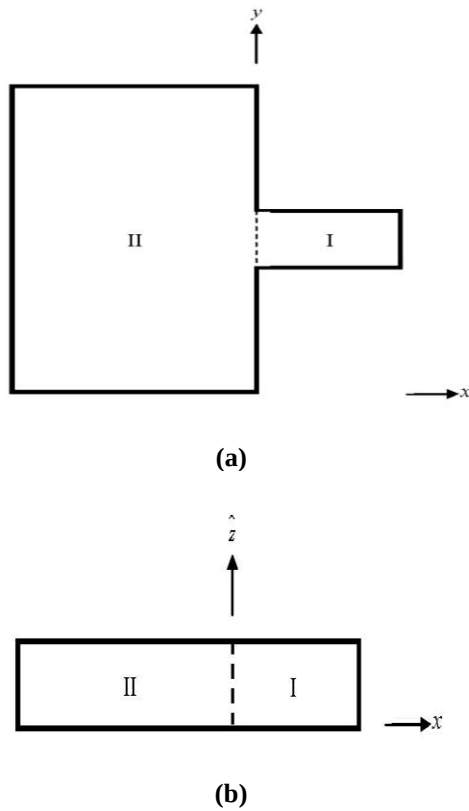


Figure 2: The divisions used for the antenna in Figure 1: (a) top view, (b) front-side view.

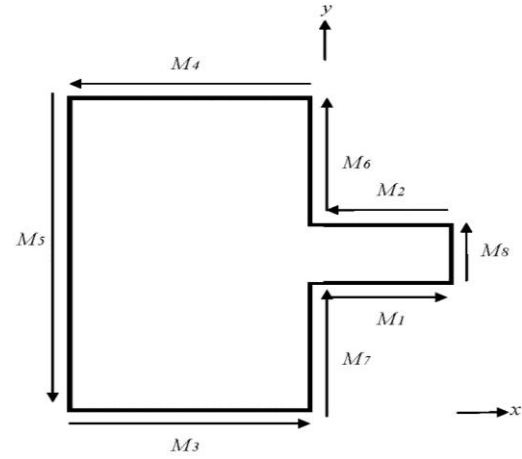
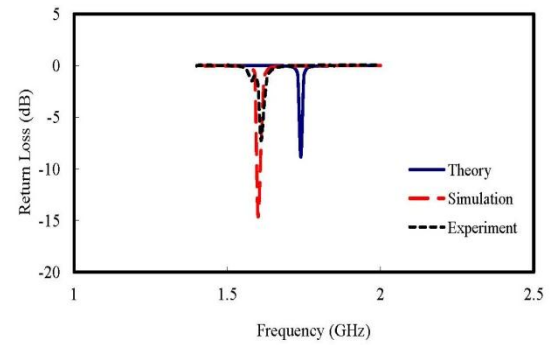
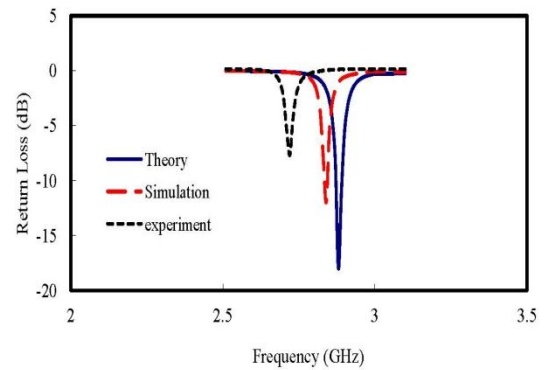


Figure 3: The magnetic currents for the antenna in Figure 1.

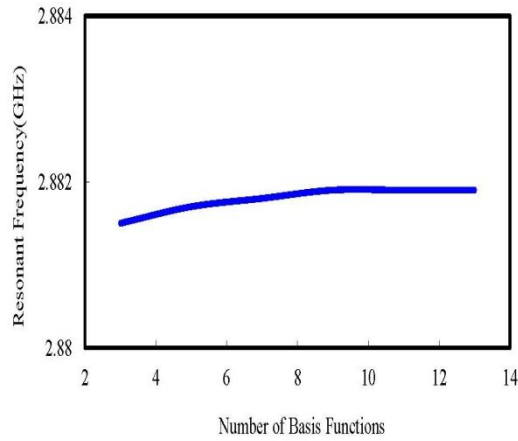


(a)

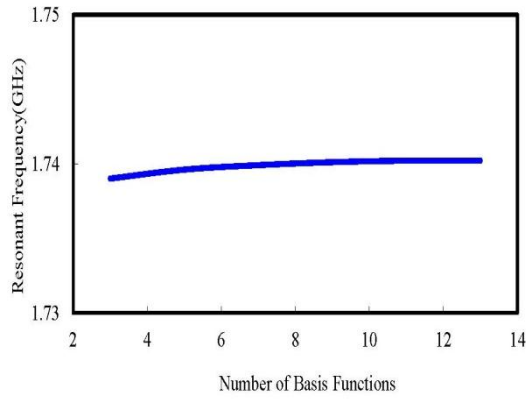


(b)

Figure 4: For the antenna in Figure 1, the S_{11} parameter obtained from theory, simulations and experiments : (a) lower mode, (b) higher mode, where the antenna dimensions : $b=3\text{cm}$, $\kappa=2\text{ cm}$, $d=2\text{ cm}$, $l=2\text{ cm}$, $w=1\text{ cm}$, $t=20\text{ mil}$, $\epsilon_r=3.35$, and feeding at (2.5,2.5).

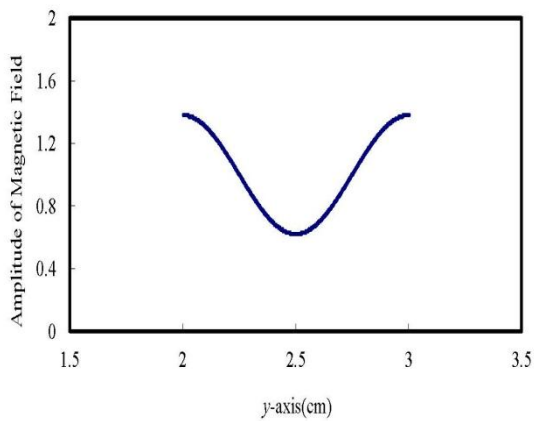


(a)

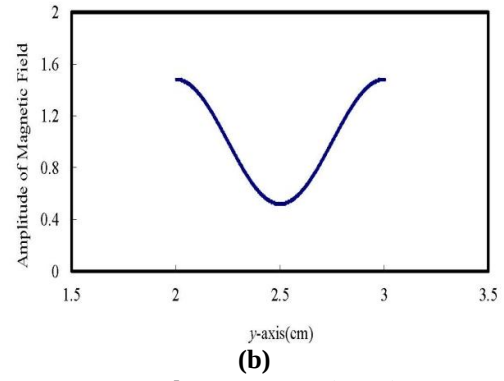


(b)

Figure 5: For the antenna in Figure 1, the theoretical resonant frequency vs. the number of chosen basis functions : (a) lower mode, (b) higher mode, where the ratio of harmonics and chosen basis functions is 50, and the dimensions used are the same as those in Figure 4.

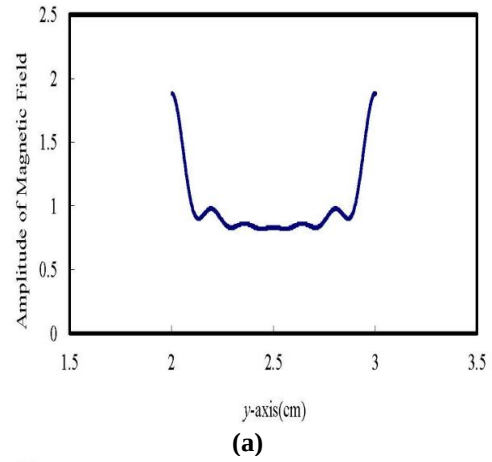


(a)

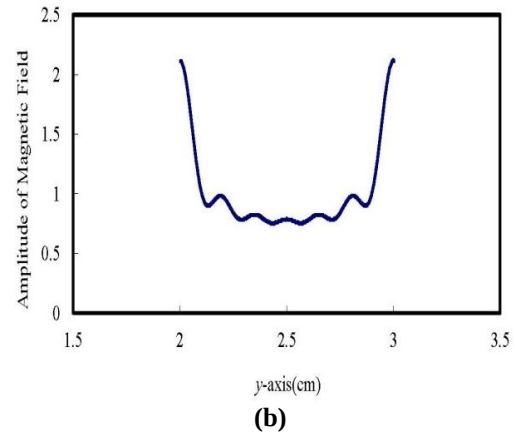


(b)

Figure 6: For the antenna in Figure 1, the theoretical distribution of magnetic fields at the conjunction : (a) lower mode, (b) higher mode, where the number of chosen basis functions is 3, the ratio of harmonics and chosen basis functions is 50, and the dimensions used are the same as those in Figure 4.

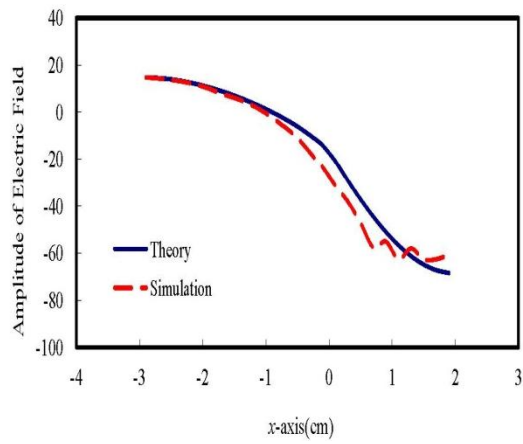


(a)

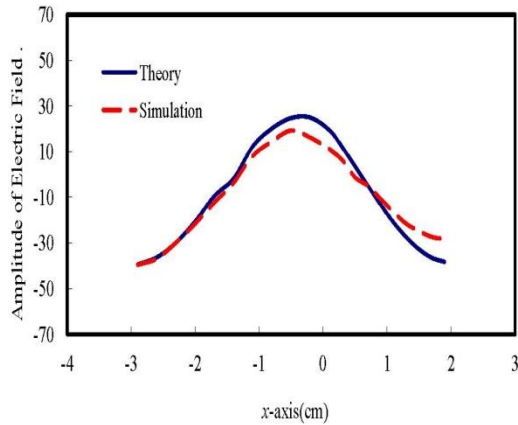


(b)

Figure 7: For the antenna in Figure 1, the theoretical distribution of magnetic fields at the conjunction : (a) lower mode, (b) higher mode, where the number of chosen basis functions is 13, the ratio of harmonics and chosen basis functions is 50, and the dimensions used are the same as those in Figure 4.

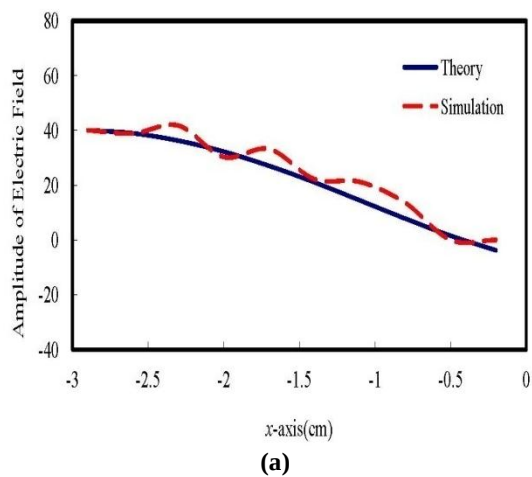


(a)

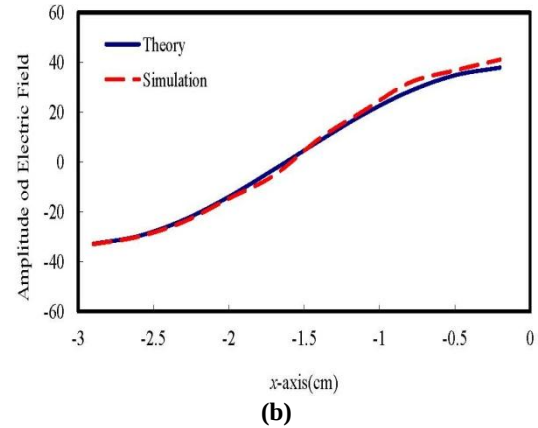


(b)

Figure 8: For the antenna in Figure 1, the theoretical and simulated distributions of electric fields along the cross section at $y=c+w/2$: (a) lower mode, (b) higher mode, where the number of chosen basis functions is 13, the ratio of harmonics and chosen basis functions is 50, and the dimensions used are the same as those in Figure 4.

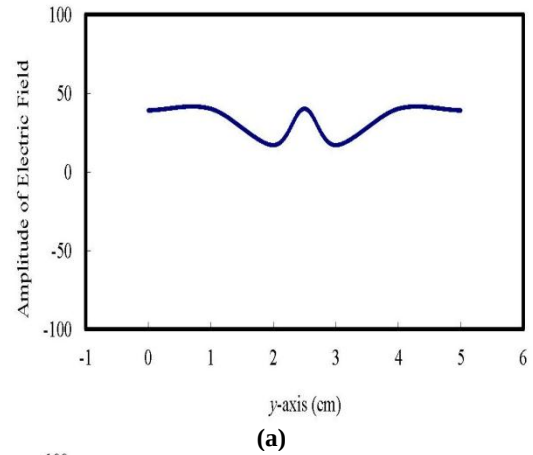


(a)

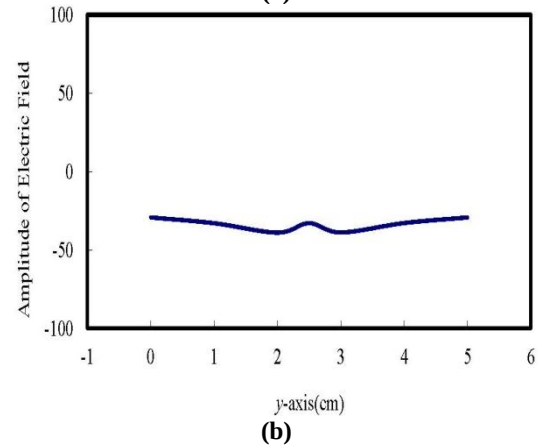


(b)

Figure 9: For the antenna in Figure 1, the theoretical and simulated distributions of electric fields along the cross section at $y=c+d+w/2$: (a) lower mode, (b) higher mode, where the number of chosen basis functions is 13, the ratio of harmonics and chosen basis functions is 50, and the dimensions used are the same as those in Figure 4.



(a)



(b)

Figure 10: For the antenna in Figure 1, the theoretical and simulated distributions of electric fields along the cross section at $x=-b$: (a) lower mode, (b) higher mode, where the number of chosen basis functions is 13, the ratio of harmonics and chosen basis functions is 50, and the dimensions used are the same as those in Figure 4.

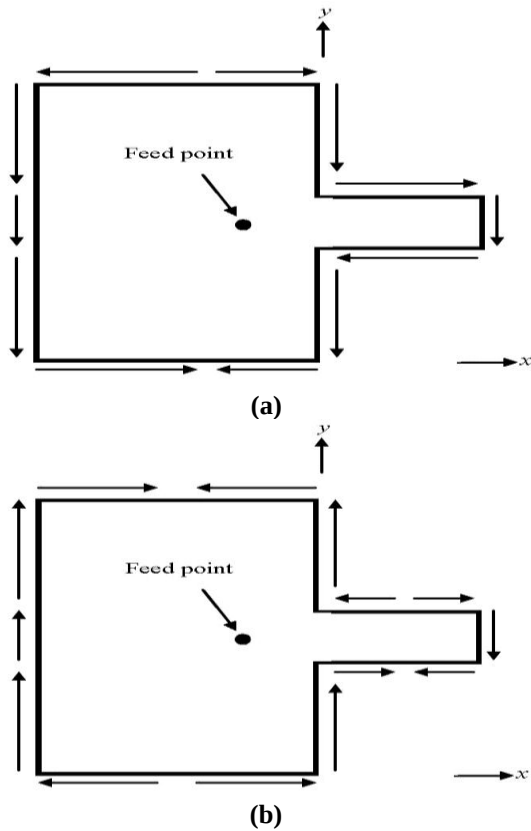


Figure 11: For the antenna in Figure 1, the corresponding magnetic currents obtained from the electric fields at various PMC walls : (a) lower mode, (b) higher mode, and the dimensions used are the same as those in Figure 4.

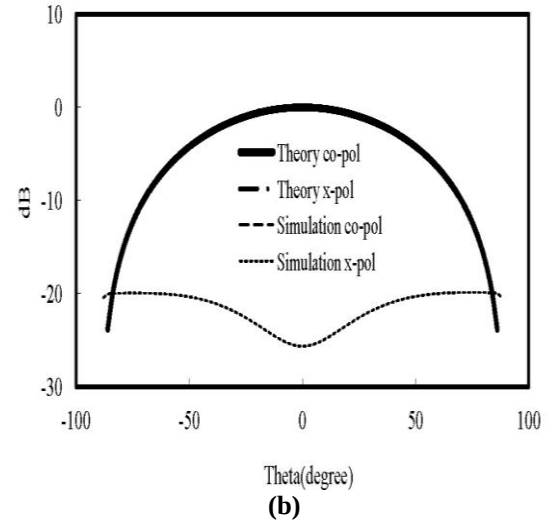
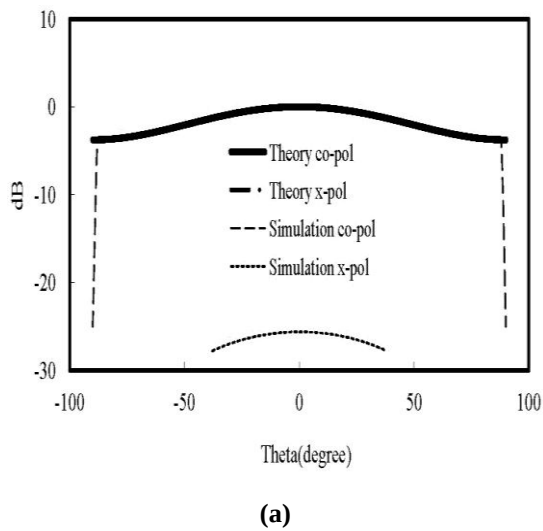


Figure 12: For the lower mode of the antenna in Figure 1, the theoretical and simulated radiation patterns on various planes : (a) $\phi=0^\circ$, (b) $\phi=90^\circ$, and the dimensions used are the same as those in Figure 4.

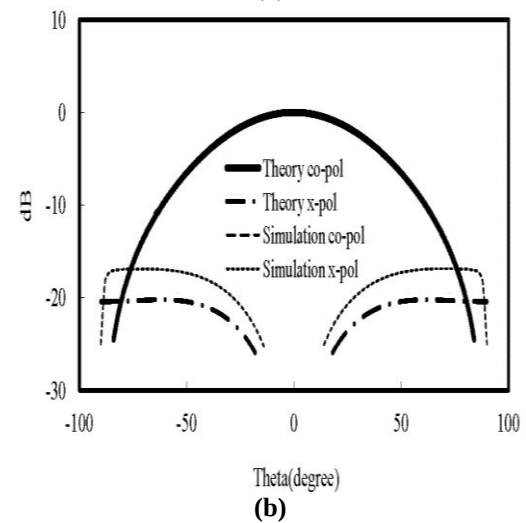
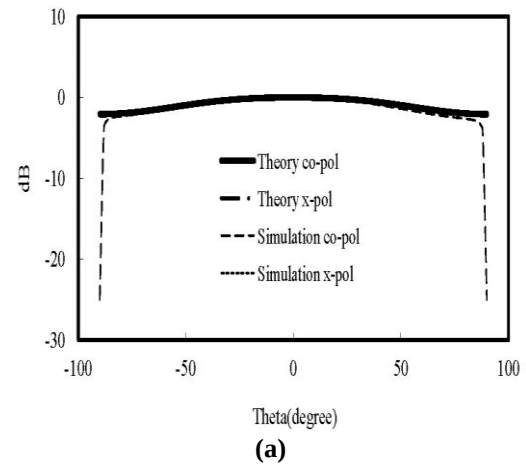


Figure 13: For the higher mode of the antenna in Figure 1, the theoretical and simulated radiation patterns on various planes : (a) $\phi=0^\circ$, (b) $\phi=90^\circ$, and the dimensions used are the same as those in Figure 4.

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